LECTURE 10:
CONDITIONAL EXPECTATION

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Our next aim is to study *stochastic processes*; that is, sequences of random variables that have nontrivial relationships between each other that we are specifically interested in. Such sequences are often viewed as an evolution of a random variable through time:

Definition 10.1. A **stochastic process** is a map $X : \mathcal{T} \times \Omega \to \mathcal{S}$ defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and *time index* $\mathcal{T}$ that maps to a *state space* $\mathcal{S}$. Here, $\mathcal{S}$ is a measure space, and $\mathcal{T}$ can be discrete, countable, or continuous (e.g., $\{0, 1, \ldots, n\}, \mathbb{N}, \mathbb{Z}, [0, T]$).

Depending on the context, a stochastic process can be considered

- *as a collection of random variables*: for any $t \in \mathcal{T} : X(t, .) : \Omega \to \mathcal{S}$ is a $\mathcal{T}$-indexed random variable; or
- *as a random trajectory*: for any $\omega \in \Omega : X(., \omega) : \mathcal{T} \to \mathcal{S}$ is called a sample path.

For example, in the discrete case when $\mathcal{T} = \{0, 1, 2, \ldots\}$ (which is the main case that we will study in this class), we can interpret X

- *as a collection of RVs*: $X_0, X_1, X_2, \ldots$; or
- *as a random trajectory*: $\{X_i(\omega)\}_{i=1}^{\infty}$ is a sequence of numbers, which is itself a random function in the sense that each $\omega \in \Omega$ corresponds to a possibly different sequence.

There are many types of stochastic processes, motivated by various applications and emphasizing different properties of how randomness evolves over time. In this class, we will have time to look closer on two types of processes, informally described as follows:

- **Martingales**: in expectation, the process does not change with time,
- **Markov chains**: conditioned on the present, the past and future are independent.

So, we need to have a formal framework for the dependence (conditioning) of some random variables on other random variables. Recall from undergraduate probability that for events A, B, the conditional probability of A given B is defined by $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$, which is well-defined whenever $\mathbb{P}(B) > 0$. When X and Y are random variables, what do we mean by $(X \mid Y)$? How do we define $\mathbb{E}(X \mid Y)$ more generally?

Example. If X and Y are discrete random variables defined on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$ that take *finitely many values*, the extension from the definition of conditional

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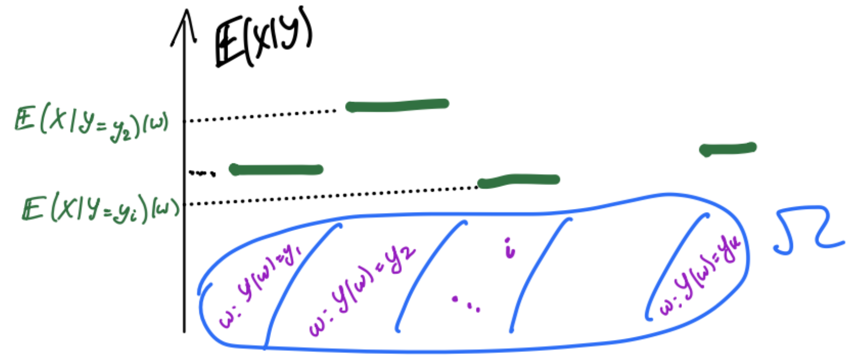
probabilities for events is clear. Namely, if $X \in \{x_1, \dots, x_k\}$ and $Y \in \{y_1, \dots, y_m\}$, then for every $i \in \{1, 2, \dots, m\}$,

$$\mathbb{E}[X \mid Y = y_i] = \sum_{j=1}^{k} x_j \mathbb{P}(X = x_j \mid Y = y_i).$$

Note that this defines a new random variable $\bar{X}$ on the same space $(\Omega, \mathcal{F}, \mathbb{P})$:

$$\bar{X} : \omega \mapsto \sum_{j=1}^{k} x_j \mathbb{P}(X = x_j \mid Y = y_i) \text{ for all } \omega \text{ such that } Y(\omega) = y_i.$$

We denote this as $\mathbb{E}[X \mid Y] := \bar{X}$. *Question:* how do we extend this idea to general X, Y?



Within the same example, observe that:

(1) The σ-algebra generated by Y is $\sigma(Y) = \sigma(\{Y = y_1\}, \dots, \{Y = y_m\})$. Thus, $\bar{X}$ is constant on every generator set of the form $\{Y = y_i\}$. So, the preimage of every Borel set will be a union of the generators, and hence $\bar{X}$ is measurable with respect to $\sigma(Y)$ (which did not have to be the case for X itself!).
(2) The following calculation holds:

$$\begin{aligned}\mathbb{E}[\bar{X}\mathbb{1}_{\{Y=y_i\}}] &= \sum_{j=1}^{k} x_j \mathbb{P}(X = x_j \mid Y = y_i)\mathbb{P}(Y = y_i) \\ &= \sum_{j=1}^{k} x_j \mathbb{P}(X = x_j, Y = y_i) = \mathbb{E}[X\mathbb{1}_{\{Y=y_i\}}].\end{aligned}$$

To summarize, we observe that the conditional expectation of X with respect to Y is a "version" of X that is "better-behaved with respect to Y" (measurable with respect to $\sigma(Y)$) and has the same expectations as X on sets that are determined by Y (i.e., on sets in $\sigma(Y)$). These properties motivate the following general definition:

Definition 10.2 (Conditional expectation). Let X and Y be integrable random variables on the same probability space. We say that a random variable $\bar{X}$ is a (version of the) *conditional expectation of X given Y*, and we write $\mathbb{E}[X \mid Y] := \bar{X}$, if

- $\bar{X}$ is $\sigma(Y)$-measurable, and
- $\mathbb{E}[\bar{X}\mathbb{1}_A] = \mathbb{E}[X\mathbb{1}_A]$ for any $A \in \sigma(Y)$.

In general, we can also define the expectation conditional on a σ-algebra:

Definition 10.3 (**Conditional expectation with respect to a σ-algebra**)**.** Let X be an integrable random variable on $(\Omega, \mathcal{F}, \mathbb{P})$. For any sub-$\sigma$-algebra $\mathcal{G} \subseteq \mathcal{F}$, we say that a $\mathcal{G}$-measurable random variable $\bar{X}$ is a *conditional expectation of* X *given* $\mathcal{G}$ if $\mathbb{E}[\bar{X}\mathbb{1}_A] = \mathbb{E}[X\mathbb{1}_A]$ for any $A \in \mathcal{G}$.

Key question: The definition above does not provide an explicit construction of a conditional expectation. Why does such a random variable exist, and why is it well-defined (i.e., unique, in an almost sure sense, with the required properties)?

Existence and uniqueness

Uniqueness. Suppose that $\bar{X}$ and $\hat{X}$ are are both conditional expectations given a sub-σ-algebra $\mathcal{G}$. Then for any $A \in \mathcal{G}$,

$$\mathbb{E}[\bar{X}\mathbb{1}_A] = \mathbb{E}[X\mathbb{1}_A] = \mathbb{E}[\hat{X}\mathbb{1}_A].$$

Consider $A_n := \{\omega \in \Omega : \bar{X}(\omega) - \hat{X}(\omega) \geq n^{-1}\}$. For any $n \in \mathbb{Z}_+$, $A_n \in \mathcal{G}$ since $\bar{X}$ and $\hat{X}$ are $\mathcal{G}$-measurable. Then,

$$0 = \mathbb{E}[(\bar{X} - \hat{X})\mathbb{1}_{A_n}] \geq \frac{1}{n}\mathbb{P}(A_n),$$

which shows that $\mathbb{P}(A_n) = 0$ for every integer n, and their union also has measure zero:

$$\mathbb{P}(\cup_{n=1}^{\infty} A_n) = \mathbb{P}\{\bar{X} > \hat{X}\} = 0.$$

Analogously, $\mathbb{P}\{\bar{X} < \hat{X}\} = 0$, so we conclude that $\bar{X} = \hat{X}$ almost surely.

Existence. To construct the conditional expectation of X given the sub-σ-algebra $\mathcal{G}$, recall that we can write for any $A \in \mathcal{G}$,

$$\nu(A) := \int_A X \, \mathrm{d}\mathbb{P} = \mathbb{E}[X\mathbb{1}_A].$$

We assume that $X \geq 0$. (For general X, we consider its positive and negative parts separately and apply the same argument.) Then, ν is a finite measure since $\nu(\Omega) = \mathbb{E}(X) < \infty$.

We want to construct a $\mathcal{G}$-measurable random variable $\bar{X}$ with the same integral over every $A \in \mathcal{G}$. *Sanity check:* X itself does not work since it could be not measurable with respect to $\mathcal{G}$ (even though we can integrate it over every A in the sub-sigma-algebra). To restate the goal: we need to represent the measure $\nu(A) = \mathbb{E}[\bar{X}\mathbb{1}_A]$ on $\mathcal{G}$; that is, $\nu(A) = \int_A \bar{X} \, \mathrm{d}\mu$ for μ being the restriction of $\mathbb{P}$ to $\mathcal{G}$.[1]

This is actually a non-trivial result. A general reason why this is possible uses the following result from measure theory:

Theorem 10.4 (Radon-Nikodym theorem)**.** *Let μ_1 and μ_2 be two σ-finite measures on a measure space $(\Omega, \mathcal{F})$. Suppose that μ_1 is absolutely continuous with respect to μ_2, denoted as $\mu_1 \ll \mu_2$, which means that for any $A \in \mathcal{F}$ such that $\mu_2(A) = 0$, we have $\mu_1(A) = 0$. Then there exists a non-negative $\mathcal{F}$-measurable function $f : \Omega \to [0, \infty)$ such that for any $A \in \mathcal{F}$,*

$$\mu_1(A) = \int_A f \, \mathrm{d}\mu_2.$$

[1]This is a probability measure on $(\Omega, \mathcal{G})$ such that $\mu(A) = \mathbb{P}(A)$ for any $A \in \mathcal{G}$

We won't do the proof of Theorem 10.4 here (see, e.g., [Dembo, Section 4.1.2]). To get more comfortable with the statement, observe that

- A Gaussian random variable is absolutely continuous with respect to the Lebesgue measure on $\mathbb{R}$. More generally, any random variable with a density is absolutely continuous with respect to the Lebesgue measure.
- Discrete random variables with the same support are absolutely continuous with respect to each other.

For our setting, where $\mu_1 = \mathbb{P}$ and $\mu_2 = \nu$, we have, by construction, $\nu \ll \mathbb{P}$: integrating over a set of measure zero gives zero! So, by Theorem 10.4, there exists a $\mathcal{G}$-measurable RV $\bar{X}$ that satisfies

$$\mathbb{E}[X\mathbb{1}_A] = \nu(A) = \int_A \bar{X}\, \mathrm{d}\mathbb{P} = \mathbb{E}[\bar{X}\mathbb{1}_A].$$

To summarize, we define the (almost surely unique) conditional expectation of X given $\mathcal{G}$ by $E[X \mid \mathcal{G}] := \bar{X}$.

Properties of conditional expectation

We will discuss some intuition behind the meaning of conditional expectations:

(a) Conditioned on an event, $\mathbb{E}[X \mid A]$: the additional information learned from the experiment is whether $\omega \in A$ or not, and we recalculate the expectation according to $\mathbb{P}(\cdot \mid A)$;
(b) Conditioned on a discrete random variable, $\mathbb{E}[X \mid Y]$: the additional information on ω is which set $\{Y = y_j\}$ it is in, so $\mathbb{E}[X \mid Y](\omega)$ is a random variable that is constant on each of these sets (calculated as in (a) on each set);
(c) Conditioned on a sub-σ-algebra, $\mathbb{E}[X \mid \mathcal{G}]$: the additional information on ω is $\{Y(\omega) : Y \text{ is } \mathcal{G}\text{-measurable}\}$, so $\mathbb{E}[X \mid \mathcal{G}](\omega)$ is a random variable averaging X given this information.

The larger $\mathcal{G}$ is, the more we "know" about X, and so $\mathbb{E}[X \mid \mathcal{G}]$ is able to more accurately describe X. Consider two extreme cases. If $\mathcal{G} = \{\emptyset, \Omega\}$ is the trivial σ-algebra (i.e., no knowledge), then $\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X]$ is the usual expectation (our best guess of X is a single value that is its expectation). If $\mathcal{G} = \mathcal{F}$ (i.e., everything is known), then $\mathbb{E}[X \mid \mathcal{G}] = X$ (we know the exact value of X itself).

The following summarizes many useful properties of conditional expectations:

Theorem 10.5 (Properties of conditional expectations)**.** *Let X, X_1, X_2 be integrable random variables on $(\Omega, \mathcal{F}, \mathbb{P})$, and $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-algebra.*

(1) *If X is $\mathcal{G}$-measurable, then* $\mathbb{E}[X \mid \mathcal{G}] = X$
(2) ***Linearity:*** *for all* $\alpha, \beta \in \mathbb{R}$, $\mathbb{E}[\alpha X_1 + \beta X_2 \mid \mathcal{G}] = \alpha\mathbb{E}[X_1 \mid \mathcal{G}] + \beta\mathbb{E}[X_2 \mid \mathcal{G}]$.
(3) ***Monotonicity:*** *if $X_1 \le X_2$ a.s., then* $\mathbb{E}[X_1 \mid \mathcal{G}] \le \mathbb{E}[X_2 \mid \mathcal{G}]$.
(4) ***Expectations agree:*** $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X]$.
(5) ***Tower property:*** *If $\mathcal{H} \subseteq \mathcal{G} \subseteq \mathcal{F}$ are sub-σ-algebras, then*

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{H}] \mid \mathcal{G}] = \mathbb{E}[X \mid \mathcal{H}] \quad a.s.$$

Proof. (1) is immediate by definition; think it through!

(2) is a direct check by definition: for any $A \in \mathcal{G}$,

$$\begin{aligned}\mathbb{E}[\mathbb{1}_A(\alpha X_1 + \beta X_2)] = \alpha\mathbb{E}[\mathbb{1}_A X_1] + \beta\mathbb{E}[\mathbb{1}_A X_2] &= \alpha\mathbb{E}[\mathbb{1}_A\mathbb{E}[X_1 \mid \mathcal{G}]] + \beta\mathbb{E}[\mathbb{1}_A\mathbb{E}[X_2 \mid \mathcal{G}]] \\ &= \mathbb{E}[\mathbb{1}_A(\alpha\mathbb{E}[X_1 \mid \mathcal{G}] + \beta\mathbb{E}[X_2 \mid \mathcal{G}])].\end{aligned}$$

(3) It is enough to show that if $X \geq 0$ then $\bar{X} := \mathbb{E}[X \mid \mathcal{G}] \geq 0$. To do this, consider

$$A := \{\omega \in \Omega : \bar{X} \leq 0\} \in \mathcal{G}.$$

Then $0 \geq \mathbb{E}[\bar{X}\mathbb{1}_{\{\bar{X}\leq 0\}}] = \mathbb{E}[X\mathbb{1}_{\{\bar{X}\leq 0\}}] \geq 0$, and so $\bar{X} \geq 0$ almost surely.

(4) $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]\mathbb{1}_\Omega] = \mathbb{E}[X\mathbb{1}_\Omega] = \mathbb{E}[X]$.

(5)
- $\bar{X} := \mathbb{E}[X \mid \mathcal{H}]$ is $\mathcal{H}$-measurable and thus also $\mathcal{G}$-measurable, so $\mathbb{E}[\bar{X} \mid \mathcal{G}] = \bar{X}$ by property (1).
- To show $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] = \bar{X}$, we need to show that for any $A \in \mathcal{H}$,

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]\mathbb{1}_A] = \mathbb{E}[\bar{X}\mathbb{1}_A].$$

Indeed, for any $A \in \mathcal{H} \subseteq \mathcal{G}$,

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]\mathbb{1}_A] = \mathbb{E}[X\mathbb{1}_A] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{H}]\mathbb{1}_A].$$

□

Next, we prove three more properties that require a little more care:

Theorem 10.6 (Properties of conditional expectations II)**.** *Let X be an integrable random variable on $(\Omega, \mathcal{F}, \mathbb{P})$, and $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-algebra.*

(i) ***Conditional MCT:*** *If $X_n \geq 0$ and $X_n \to X$ monotonically increasing a.s., then*

$$\mathbb{E}[X_n \mid \mathcal{G}] \to \mathbb{E}[X \mid \mathcal{G}] \quad a.s.$$

(ii) ***Factoring out "the known":*** *If Z is a $\mathcal{G}$-measurable random variable on the same space $(\Omega, \mathcal{F}, \mathbb{P})$ (which is a non-trivial additional condition if $\mathcal{G}$ is smaller than $\mathcal{F}$), then*

$$\mathbb{E}[ZX \mid \mathcal{G}] = Z\mathbb{E}[X \mid \mathcal{G}] a.s.$$

(iii) ***Independence:*** *If X is independent from $\mathcal{G}$ (that is, $\sigma(X)$ and $\mathcal{G}$ are independent), then*

$$\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X] \quad a.s.$$

Moreover, for any sub-σ-algebra $\mathcal{H} \subseteq \mathcal{F}$ such that $\mathcal{G}$ is independent from $\sigma(X, \mathcal{H})$,

$$\mathbb{E}[X \mid \sigma(\mathcal{G}, \mathcal{H})] = \mathbb{E}[X \mid \mathcal{H}] \quad a.s.$$

Proof. (i) Denote $Y_n := \mathbb{E}[X_n \mid \mathcal{G}]$. By monotonicity of conditional expectations, $\{Y_n\}_n$ is an increasing sequence that has a pointwise limit which we will denote by Y. We need to show that $Y = \mathbb{E}[X \mid \mathcal{G}]$. Indeed, Y is $\mathcal{G}$-measurable as a pointwise limit of $\mathcal{G}$-measurable functions, and for any $A \in \mathcal{G}$, $\mathbb{E}[Y\mathbb{1}_A] = \mathbb{E}[X\mathbb{1}_A]$. The latter claim follows from

$$\mathbb{E}[X\mathbb{1}_A] \overset{\text{MCT}}{=} \lim_n \mathbb{E}[X_n\mathbb{1}_A] = \lim_n \mathbb{E}[Y_n\mathbb{1}_A] \overset{\text{MCT}}{=} \mathbb{E}[Y\mathbb{1}_A].$$

(ii) By linearity, we can begin by assuming that $X \geq 0$. Then, we follow the standard 4-step argument. Step 1: If $Z = \mathbb{1}_A$, $A \in \mathcal{G}$, then we need to show that for any $B \in \mathcal{G}$,

$$\mathbb{E}[X\mathbb{1}_A\mathbb{1}_B] = \mathbb{E}[\mathbb{1}_A\mathbb{E}[X \mid \mathcal{G}]\mathbb{1}_B].$$

This holds by definition of conditional expectation since $\mathbb{1}_A \mathbb{1}_B = \mathbb{1}_{A\cap B}$ is $\mathcal{G}$-measurable. Step 2: For non-negative simple functions, the statement follows from linearity. Step 3. For non-negative measurable functions Z, we can apply the monotone convergence theorem. Step 4. By linearity, we conclude for arbitrary (integrable) Z with $Z = Z_+ - Z_-$.

(iii) Denote $\bar{X} := \mathbb{E}[X \mid \mathcal{H}]$. We need to show that $\mathbb{E}[X\mathbb{1}_A] = \mathbb{E}[\bar{X}\mathbb{1}_A]$ for any $A \in \sigma(\mathcal{G}, \mathcal{H})$. By Dynkin's extension theorem, it is enough to show this for a π-system generating $\sigma(\mathcal{G}, \mathcal{H})$; namely, it is enough to take $A = G \cap H$ for all $G \in \mathcal{G}$ and $H \in \mathcal{H}$. Indeed, by independence, we can split the expectation into a product of expectations:

$$\mathbb{E}[X\mathbb{1}_{H\cap G}] = \mathbb{E}[X\mathbb{1}_H\mathbb{1}_G] = \mathbb{E}[X\mathbb{1}_H]\mathbb{E}[\mathbb{1}_G] = \mathbb{E}[\bar{X}\mathbb{1}_H]\mathbb{E}[\mathbb{1}_G] = \mathbb{E}[\bar{X}\mathbb{1}_{H\cap G}].$$

□

Remark 10.7. Similarly to the monotone convergence theorem, other standard facts about expectations also extend to the conditional expectation, including the following:

- **Conditional Jensen's inequality:** If $g : \mathbb{R} \to \mathbb{R}$ is convex and $g(X)$ is integrable, then
$$\mathbb{E}[g(X) \mid \mathcal{G}] \geq g(\mathbb{E}[X \mid \mathcal{G}]) \quad \text{a.s.}$$
- **Conditional DCT:** If $|X_n| \leq Y$ for an integrable random variable Y and $X_n \to X$, then
$$\mathbb{E}[X_n \mid \mathcal{G}] \to \mathbb{E}[X \mid \mathcal{G}] \quad \text{a.s.}$$
- **Conditional Fatou's lemma:** If $X_n \geq 0$, then
$$\mathbb{E}[\liminf X_n \mid \mathcal{G}] \leq \liminf \mathbb{E}[X_n \mid \mathcal{G}] \quad \text{a.s.}$$

The proofs of these results are left as exercises.

Geometric proof of existence

Another way to construct the conditional expectation is by constructing *an orthogonal projection*. This works nicely in the space of square integrable random variables: if $X, Y \in L^2(\Omega, \mathcal{F}, \mathbb{P})$, then we can define

- the *scalar product* between them by $\langle X, Y\rangle := \mathbb{E}[X \cdot Y] < \infty$ (by Cauchy-Schwarz);
- the *norm* $\|X\|_2 := \sqrt{\mathbb{E}[X^2]} < \infty$; and
- *orthogonality* by saying that $X \perp Y$ if $\mathbb{E}[X \cdot Y] = 0$.

Theorem 10.8. *Let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-algebra. For every random variable $X \in L^2(\Omega, \mathcal{F}, \mathbb{P})$, there is a unique (up to changes of measure zero) random variable $Y \in L^2(\Omega, \mathcal{G}, \mathbb{P})$, which is the orthogonal projection of X onto $L^2(\Omega, \mathcal{G}, \mathbb{P})$ such that*

(a) *Y is closest to X in $L^2(\Omega, \mathcal{G}, \mathbb{P})$; that is*
$$\|X - Y\|_2 = \inf\{\|X - Z\|_2 : Z \in L^2(\Omega, \mathcal{G}, \mathbb{P})\}.$$
(b) *For any $Z \in L^2(\Omega, \mathcal{G}, \mathbb{P})$, we have $\langle X - Y, Z\rangle = 0$.*

Once we have Theorem 10.8, we can define $\mathbb{E}[X \mid \mathcal{G}] := Y$ to be the orthogonal projection of any $X \in L^2(\Omega, \mathcal{F}, \mathbb{P})$ onto $L^2(\Omega, \mathcal{G}, \mathbb{P})$. Then $\mathcal{G}$-measurability of Y follows by definition, and $\mathbb{E}[(X - Y) \cdot \mathbb{1}_A] = 0$ for any $A \in \mathcal{G}$ from (b). To finish the construction of the conditional expectation beyond the L^2 space to the class of integrable, non-negative X, consider the sequence of truncations $X_n = \min\{X, n\} \in L^2(\Omega, \mathcal{F}, \mathbb{P})$. Then use the regular monotone convergence theorem, similar to the proof of Theorem 10.6 (i). We will need linearity and monotonicity of the projection map to pass to the limits correctly. Finally, we can conclude for arbitrary X by the usual decomposition $X = X_+ - X_-$.

Proof of Theorem 10.8. We denote $L' = L^2(\Omega, \mathcal{G}, \mathbb{P})$, which is a complete subspace of $L^2(\Omega, \mathcal{F}, \mathbb{P})$, i.e., any Cauchy sequence in L' converges in L'. (The proof actually shows that it is enough to require that L' is any complete subspace of $L^2(\Omega, \mathcal{F}, \mathbb{P})$.) Note that uniqueness of the conditional expectation was already proved earlier, so we will focus on proving existence of the random variable $Y \in L^2(\Omega, \mathcal{F}, \mathbb{P})$ that satisfies properties (a) and (b).

(a): Note that we can always choose a sequence $\{Y_n\}_{n\geq 1}$, $Y_n \in L'$, that approximates the infimum: i.e., such that

$$\|X - Y_n\|_2 \to \inf\{\|X - Z\|_2 : Z \in L'\} =: I \quad \text{as } n \to \infty.$$

Next, we will show that the sequence Y_n itself converges (in L'). This follows since Y_n is a Cauchy sequence in the complete space L'. Indeed, by the parallelogram law,

$$\begin{aligned} \|X - Y_n\|_2^2 + \|X - Y_m\|_2^2 &= 2\left(\left\|X - \frac{Y_n + Y_m}{2}\right\|_2^2 + \left\|\frac{Y_n - Y_m}{2}\right\|_2^2\right) \\ &\geq 2I + \frac{1}{4}\|Y_n - Y_m\|_2^2. \end{aligned}$$

So, taking the limit of both sides as $n, m \to \infty$, we have

$$2I \geq 2I + \limsup_{m,n} \|Y_n - Y_m\|_2^2.$$

Hence, $\{Y_n\}$ is a convergent sequence. To conclude, we define Y as the limit of this sequence.

(b): We have just proved that for any fixed $Z \in L'$ and any $t \in \mathbb{R}$, we have

$$\|X - (Y + tZ)\|_2 \geq \|X - Y\|_2.$$

By expanding the square, this implies that

$$\|X - Y\|_2^2 + t^2\|Z\|_2^2 - 2t\langle X - Y, Z\rangle \geq \|X - Y\|_2^2.$$

So,

$$t\|Z\|_2^2 \geq 2\langle X - Y, Z\rangle \text{ for any } t > 0, \quad \text{and} \quad t\|Z\|_2^2 \leq 2\langle X - Y, Z\rangle \text{ for any } t < 0,$$

which allows us to conclude that $\langle X - Y, Z\rangle = 0$. ∎